FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
PART TEST – III
PAPER –2
TEST DATE: 24-12-2023

ANSWERS, HINTS & SOLUTIONS

Physics PART – I

SECTION – A

1. B

Sol. Focal length of mirror does not depend on medium.

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}+\frac{1}{(-20)}=\frac{1}{-15}$$

$\Rightarrow$ v = –60 cm

$$m=-\frac{v}{u}=-3$$

2. C

Sol. average energy density = $\frac{B_0^2}{2\mu_0}$

$$\frac{8000}{\pi}=\frac{B_0^2}{2\mu_0}$$

$B_0 = 0.08T$ …(i)

$\frac{E_0}{B_0}=c \Rightarrow E_0 = 2.4 \times 10^7$ V/m

$\omega = 2\pi f = 3.6\pi \times 10^{11}$ …(ii)

$k=\frac{\omega}{c}=\frac{3.6\pi\times10^{11}}{3\times10^8}=1.2\pi\times10^3$ …(iii)

So, $\vec{E}=2.4\times10^7 \sin\left[(3.6\pi\times10^{11})t-(1.2\pi\times10^3)y\right](-\hat{i})$ V/m and

$\vec{B}=0.08\sin\left[(3.6\pi\times10^{11})t-(1.2\pi\times10^3)y\right](+\hat{k})$ Tesla

3. C

Sol. $v_1 = \sqrt{\frac{\gamma RT_0}{M_1}}, \quad M_1 = \frac{\gamma RT_0}{v_1^2}$ …(i)

$v_2 = \sqrt{\frac{\gamma RT_0}{M_2}}, \quad M_2 = \frac{\gamma RT_0}{v_2^2}$ …(ii)

$v_{mix} = \sqrt{\frac{\gamma RT_0}{M_{mix}}}, \quad M_{mix} = \frac{\gamma RT_0}{v_{mix}^2}$ …(iii)

Molecular weight of mixture

$M_{mix} = \frac{m_1 + m_2}{n_1 + n_2} = \frac{m_1 + m_2}{\frac{m_1}{M_1} + \frac{m_2}{M_2}}$ …(iv)

From equation (i), (ii), (iii) and (iv)

$$v_{mix} = \sqrt{\frac{m_1 v_1^2 + m_2 v_2^2}{m_1 + m_2}} = 100\sqrt{\frac{152}{13}}$$

4. D

Sol. Maximum kinetic energy of α-particle is equal to Δmc^2

$\Delta m = m_U - m_{Th} - m_{He}$

= 0.00456 amu

= 7.57×10^{-30} kg

So, $(\Delta m)c^2 = 6.8 \times 10^{-13}$ J

5. C

Sol. Particles have a phase difference either zero or π.

Total mechanical energy of string remain constant.

Potential energy depends on $\frac{dy}{dx}$

6. A, B, C

Sol. Reflected light has less energy than incident light so it has more wavelength.

Radiation pressure = $\frac{2I}{C}$

$I = \frac{NhC}{A\Delta t\lambda}$

7. B, C

Sol. $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

$\frac{\mu_2}{v} = \frac{\mu_2 - \mu_1}{R} + \frac{\mu_1}{u}$

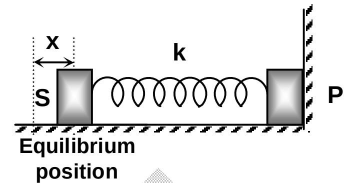
If $\mu_2 > \mu_1$, and $u > 0$ then $v > 0$

If $\mu_1 > \mu_2$, and $u < 0$ then $v < 0$

SECTION – B

8. 100

Sol. At equation



$$kx_0 = \frac{1}{4\pi\varepsilon_0}\frac{q_0^2}{L^2}$$

When block is slightly displaced.

$$f_r = \frac{q_0^2}{4\pi\varepsilon_0(L+x)^2} - k(x+x_0)$$

$$ma = -\left(k + \frac{1}{4\pi\varepsilon_0}\cdot\frac{2q_0^2}{L^3}\right)x$$

$$\omega = \sqrt{\frac{1}{m}\left(k + \frac{1}{4\pi\varepsilon_0}\frac{2q_0^2}{L^3}\right)} = 100 \text{ rad/s}$$

9. 12

Sol. When bottle is opened, volume of air is increased till pressure of 1 atm

$P_1V_1 = P_2V_2$

$3 \times 4 = 1 \times V_2 \Rightarrow V_2 = 12$ litre

So volume of air in plastic bag is 8 litre

$\Rightarrow$ Now a buoyant force is also acting on plastic bag

$F_m = v\rho g = (8 \times 10^{-3})\ (1.5) \times 10 = 120 \times 10^{-3}$ N

Required weight = 12 gm

10. 26

Sol. After refraction from AB

$$v = \frac{\mu_2}{\mu_1}u = \frac{1}{3/2}(-3) = -2 \text{ cm}$$

Now, refraction from CD

$$v = \frac{\mu_2}{\mu_1}u = \frac{-2}{1}(2+p) = -(4+2p)$$

Final refraction from curved surface

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1}{\infty} - \frac{2}{-(4+2p+m)} = \frac{1-2}{-15}$$

30 = 4 + 2p + m

m = 26 – 2p

minimum value of ‘p’ can be zero

$(m)_{max}$ = 26 cm

11. 95

Sol. $$\frac{GM_P}{\left(\frac{R_0}{3}\right)^2} = \frac{1}{4}\frac{GM_e}{R_e^2}$$

$$M_P = \frac{1}{36}M_e \qquad \ldots(i)$$

At equator

$$(g')_P = g_P - R_P\omega^2\cos^2\lambda$$

$$0 = \frac{GM_P}{(R_e/3)^2} - \left(\frac{R_e}{3}\right)\omega^2$$

$$\frac{9G\left(\frac{M_e}{36}\right)}{R_e^2} = \frac{R_e}{3}\omega^2$$

$$\omega^2 = \frac{3}{4}\frac{GM_e}{R_e^3}$$

$$\left(\frac{2\pi}{T}\right)^2 = \frac{3}{4}\frac{G\left(\rho\times\frac{4}{3}\pi R_e^3\right)}{R_e^3}$$

$$T = \sqrt{\frac{88}{7G\rho}}$$

12. 12

Sol. The path difference due to sheet-1 is

Area under the curve = $\left(\frac{1}{2}\times t\times 1\right) + (2t) = \frac{5t}{2}$

Net path difference at 'O'

$$\Delta x = \left[\mu_3(SS_2 - t_0) + \mu_2 t_0 + \mu_4(S_2O)\right] - \left[\mu_3 SS_1 + \mu_4(S_1O - t) + \frac{5t}{2}\right]$$

$$= \mu_3(SS_2 - SS_1) - \mu_3 t_0 + \mu_2 t_0 + \mu_4(S_2O - S_1O) + \mu_4 t - \frac{5t}{2}$$

For central maxima, $\Delta x = 0$

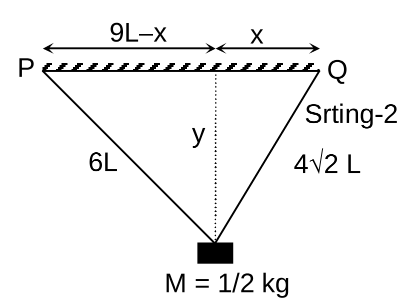
$$0 = \frac{3}{2}\left(\frac{d^2}{2D}\right) - \frac{1}{6}t_0 - \frac{1}{2}t$$

$$\frac{t}{2} + \frac{t_0}{6} = \frac{3d^2}{4D}$$

$$t = \frac{5}{12}\mu m$$

13. 4529

Sol. $(6L)^2 = (9L - x)^2 + y^2$



$36L^2 = 81L^2 + x^2 - 18Lx + y^2$ …(i)

$(4\sqrt{2}L)^2 = x^2 + y^2$

$32L^2 = x^2 + y^2$ …(ii)

From (i) and (ii)

$4L^2 = 81L^2 - 18Lx$

18x = 77 L

$x = \frac{77L}{18}$ …(iii)

So, $y^2 = 32L^2 - \left(\frac{77L}{18}\right)^2 = \frac{4439}{81}$

$$y = \frac{\sqrt{4439}}{9}$$

$$T = 2\pi\sqrt{\frac{y}{g}} = 2\pi\sqrt{\frac{\sqrt{4439}}{90}}$$

$$\lambda_1 + \lambda_2 = 4529$$

SECTION – C

14. 0.20
Range (0.20 to 0.22)

15. 0.20
Range (0.19 to 0.21)

Sol. (Q.14-15)
Torque due to force on the face ADHE is

$$d\tau_1 = \rho gxbdx(H - x)$$

$$\tau_1 = \rho gb\int_0^H (xH - x^2)dx = \rho gb\frac{H^3}{6} \quad \textbf{...(i)}$$

Torque due to force on the face ABCD is

$$\tau_2 = (\rho gH)b\ell\frac{\ell}{2} = \rho gbH\cdot\frac{\ell^2}{2} \quad ...(ii)$$

Torque due to weight Mg

$$\tau_3 = Mg\frac{\ell}{2} = \rho_b(\ell bh)g\frac{\ell}{2} = \rho_b gb\frac{\ell^2 h}{2} \quad ...(iii)$$

For no toppling about MN

$$\tau_3 > \tau_1 + \tau_2$$

$$\frac{k\ell^2 h}{2} > \frac{(4h/5)^3}{6} + \frac{4h}{5}\cdot\frac{\ell^2}{2} \quad \text{(k is R.D.)}$$

$$\left(\frac{k}{2} + \frac{2}{5}\right)\ell^2 > \frac{64}{125\times 6}h^2$$

$$\frac{\ell}{h} > \frac{4}{5\sqrt{15}}$$

$$\left(\frac{\ell}{h}\right)_{min} = 0.206$$

$$\frac{\tau_2}{\tau_1} = \frac{H\ell^2/2}{H^3/6} = \frac{25\times 3}{16}\times\frac{16}{25\times 15} = 0.2$$

16. 8897.00
Range (8895 to 8899)

17. 145.13
Range (145.10 to 145.15)

Sol. (Q.16-17):

$$T_i = \frac{2\pi(r_i)^{3/2}}{\sqrt{GM_e}} = \frac{2\pi(9\times 10^6)^{3/2}}{\sqrt{6.67\times 10^{-11}\times 6\times 10^{24}}} = 8478\text{ sec}$$

$$(E_i)_A = \frac{-GM_e m_A}{2r} = \frac{-6.67\times10^{-11}\times6\times10^{24}\times300}{2(9\times10^6)} = -667\times10^7\ \text{J}$$

$$(v_i)_A = \sqrt{\frac{GM_e}{r}} = \sqrt{\frac{6.67\times10^{-11}\times6\times10^{24}}{9\times10^6}} = 6.67\times10^3\ \text{m/s}$$

$$(v_f)_A = (6.67\times10^3)(0.994) = 6630\ \text{m/s}$$

After the rocket is fired

$$(E_f)_A = 2(E_i)_A + \frac{1}{2}mv_f^2 = -674.7\times10^7\,\text{J m/s}$$

So, $(E_f)_A = \dfrac{-Gm_A M_e}{2r_f}$

$$r_f = \frac{-Gm_A M_e}{2(E_f)_A} = 8.897\times10^6\,\text{m} = 8897\,\text{km}$$

$$T_f = \frac{2\pi(r_f)^{3/2}}{\sqrt{GM_e}} = 8332.87\ \text{sec}$$

$$\Delta T = T_i - T_f = 145.13\,\text{sec}$$

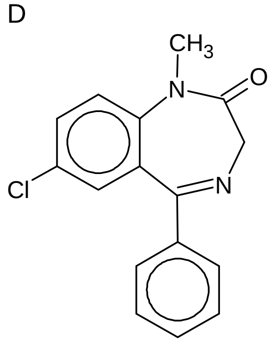
Chemistry

PART – II

SECTION – A

18. D

Sol.



(Valium)

It is a tranquiliser.

19. C

Sol. $Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O;$

$(2Cl^- \longrightarrow Cl_2 + 2e^-) \times 3$

$Cr_2O_7^{2-} + 14H^+ + 6Cl^- \longrightarrow 2Cr^{3+} + 3Cl_2 + 7H_2O;$

$E^o_{cell} = 1.23 + (-1.36)$

$= -0.13$

$$E_{cell} = -0.13 - \frac{0.0591}{6}\log\frac{[Cr^{3+}]^2[P_{Cl_2}]^3}{[Cr_2O_7^{2-}][Cl^-]^6[H^+]^{14}}$$

$$= -0.13 - \frac{0.0591}{6}\log\frac{(0.1)^2(1)^3}{(0.01)(0.1)^6(0.1)^{14}}$$

$= -0.327$ V

20. D

Sol. $NaCl + K_2Cr_2O_7 + H_2SO_4 \longrightarrow \underset{(\text{Orange red vap.})}{CrO_2Cl_2} \xrightarrow{NaCl} \underset{(\text{Yellow})}{Na_2CrO_4}$

$NaBr + K_2Cr_2O_7 + H_2SO_4 \longrightarrow Br_2 \xrightarrow{NaOH} \text{Colourless}$

21. D

Sol. Per unit cell

$N_X = \frac{7}{8}; N_Y = 3; N_Z = 1$

Hence, formula of the compound is :

$X_{7/8}Y_3Z_1 = X_7Y_{24}Z_8$

22. A, B, C

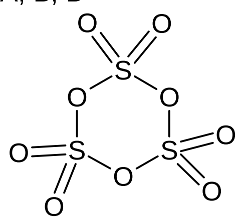
Sol. $G = f(P,T);$ $\Delta G = V\Delta P - S\Delta T$ and $\Delta T = 0; \left(\left(\frac{\partial G}{\partial P}\right)_T\right) = +V;$

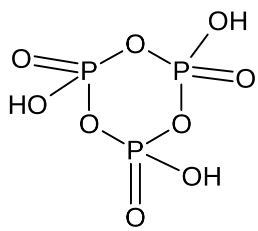
Similarly, $A = f(V,T)$; $\left(\frac{\partial A}{\partial V}\right)_T = -P$

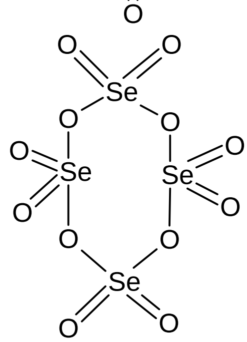
Also for isothermal process on an ideal gas $w = -q$

23. A, B, D

Sol.







24. A, B

Sol. $Cu^{2+} + CO_3^{2-}(aq) \longrightarrow CuCO_3.Cu(OH)_2 \downarrow$

(Malachite green)

$\Rightarrow$ Other three give white metal carbonates.

SECTION – B

25. 470

Sol. $\frac{r^+}{r^-} = 0.801$

Hence, AB will crystallize as BCC lattice.

$a\sqrt{3} = 2r^+ + 2r^-$

$a = \frac{814}{\sqrt{3}} = 469.98 \approx 470$ pm

26. 127

Sol. $CS_2(\ell) + 3O_2(g) \longrightarrow CO_2(g) + 2SO_2(g)$; $\Delta H = -1107.94$

$-1107.94 = (-393.5) + 2\times(-293.72) - 3\times 0 - \Delta_f H_{CS_2(\ell)}$

$\Delta_f H_{CS_2(\ell)} = 127$ kJ / mole

27. 5

Sol. Moles of HCl = mole of Na_2CO_3 (as n-factor = 1)

$= 2.1\times10^{-3}$ (in 50 ml)

Number of moles of Na_2CO_3 in 2 litre

$= \dfrac{2.1\times10^{-3}\times2\times1000}{50}$

$= 84\times10^{-3}$

$W_{Na_2CO_3} = 84\times10^{-3}\times106 = 8.9$ gm

Weight of $NaHCO_3 = 2.5$ gm

$n_{NaHCO_3} = \dfrac{2.5}{84}$

$n_{CO_2} = \dfrac{8.9}{106} + \dfrac{2.5}{84}$

Weight of $CO_2 = n_{CO_2}\times44$

= 5 gm.

28. 142

Sol. $2KClO_3 \xrightarrow[\Delta]{MnO_2} 2KCl + 3O_2$;

204.17 gm – $-(P)$

– 80 gm

$P_4 + 5O_2 \longrightarrow P_4O_{10}$;

(P) (Q)

160 gm $\longrightarrow$ 284 gm of P_4O_{10}

80 gm $\longrightarrow$ 142 gm of P_4O_{10}

29. 139

Sol. ($X = NH_3$; $Y = PH_3$; $Z = NaH_2PO_2$)

30. 287

Sol. Purple coloured complex of Co with Cl and NH_3, has the molecular formula $CoCl_3 .5NH_3$, i.e. $[Co(NH_3)_5Cl]Cl_2$. Hence, 2 mole AgCl will be precipitated on reaction with $AgNO_3$.

SECTION – C

31. 30.00

32. 82.00

Sol. As the metal gives white precipitates with NaCl and $Na_2S_2O_3$ both and white precipitate (Q) turns to black compound 'R' on standing hence the metal 'M' is 'Ag'.

$3Ag + 4HNO_3(dil.) \longrightarrow 3AgNO_3 + NO + 2H_2O$

(M) (N) (O)

$AgNO_3 + NaCl \longrightarrow AgCl$;

P(White)

$2AgNO_3 + Na_2S_2O_3 \longrightarrow Ag_2S_2O_3 \downarrow + 2NaNO_3$

Q(White)

$$2Ag_2S_2O_3 \longrightarrow \underset{(\text{Black})}{Ag_2S} + SO_3 \uparrow$$

Amount of $AgNO_3$ formed $= \frac{1}{2}$ mole.

Number of moles of $Ag_2S_2O_3$ formed $= \frac{1}{4}$ mole

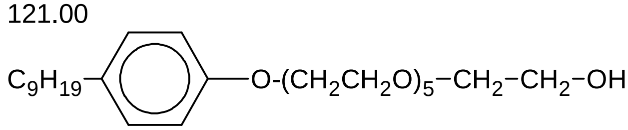
Weight of $Ag_2S_2O_3 = \frac{1}{4}(108 \times 2 + 2 \times 32 + 3 \times 16)$

= 82

33. 606.00

Sol. $(C_{17}H_{35}COO)_2$ Ca is not used in synthetic detergent.

34. 121.00

Sol. 

is used in liquid dish washing detergent.

Mathematics PART – III

SECTION – A

35. C

Sol. Let $|z + 1| = \lambda \Rightarrow \lambda \in [0, 2]$

$\Rightarrow \text{Re}(z) = \frac{\lambda^2 - 2}{2}$ and $|z^2 + 1 - z| = |\lambda^2 - 3|$

$\Rightarrow$ For $\lambda \in [0, 2]$, $f(\lambda) = \lambda + |\lambda^2 - 3| \in \left[\sqrt{3}, \frac{13}{4}\right]$

36. C

Sol. Let the two numbers be a and b

$\Rightarrow p = \frac{b + an}{n+1}$ and $q = \frac{ab(n+1)}{bn + a}$

$\Rightarrow \frac{q}{p} = \frac{(n+1)^2}{n^2 + 1 + n\left(\frac{a}{b} + \frac{b}{a}\right)}$, where $\frac{a}{b} + \frac{b}{a} \le -2$ or ≥ 2

$\Rightarrow \frac{q}{p} < 1$ or $> \left(\frac{n+1}{n-1}\right)^2$

37. B

Sol. f(x) is decreasing $\forall x \in R$ when $|p| < 3 \Rightarrow f(1) > f(2) > f(3)$

38. D

Sol. Let a_i and b_i be probabilities of i appearing on 1st and 2nd dice respectively

Then, probability of getting sum $k = \frac{1}{11}$ (where k = 2, 3, ..., 12)

$\Rightarrow \frac{1}{11} = a_1b_6 + a_2b_5 + a_3b_4 + a_4b_3 + a_5b_2 + a_6b_1$

$\Rightarrow \frac{a_1}{b_6} + \frac{a_6}{b_1} = 1$ which is not possible

39. A, B, D

Sol. $A_\lambda A_{\lambda+1} = \begin{bmatrix} C_\lambda & 0 \\ 0 & C_{\lambda-1} \end{bmatrix}\begin{bmatrix} C_{\lambda+1} & 0 \\ 0 & C_\lambda \end{bmatrix} = \begin{bmatrix} C_\lambda C_{\lambda+1} & 0 \\ 0 & C_{\lambda-1}C_\lambda \end{bmatrix}$

$\Rightarrow A_1A_2 - A_2A_3 + A_3A_4 - ..... + (-1)^{n-1} A_{n-1}A_n = \begin{bmatrix} \sum_{\lambda=1}^{n-1}(-1)^{\lambda-1} C_\lambda C_{\lambda+1} & 0 \\ 0 & \sum_{\lambda=1}^{n-1}(-1)^{\lambda-1} C_{\lambda-1}C_\lambda \end{bmatrix}$

$\Rightarrow p = \sum_{\lambda=1}^{n-1}(-1)^{\lambda-1} C_\lambda C_{\lambda+1} = {}^nC_1C_2 - {}^nC_2{}^nC_3 + {}^nC_3{}^nC_4 - ..... + (-1)^{n-2}\ {}^nC_{n-1}{}^nC_n$

$= -\left[{}^nC_0{}^nC_1 - {}^nC_1{}^nC_2 + ... + (-1)^{n-1} \cdot {}^nC_{n-1} \cdot {}^nC_n\right] + {}^nC_0{}^nC_1$

$p = -$[Coefficient of x^{n-1} in $(1 + x)^n(1 - x)^n$] + n

$$p = \begin{cases} n & ; \text{ if n is even} \\ n - (-1)^{\frac{n-1}{2}} \cdot {}^nC_{\frac{n-1}{2}} & ; \text{ if n is odd} \end{cases}$$

Similarly, $q = \sum_{\lambda=1}^{n-1} (-1)^{\lambda-1} C_{\lambda-1} C_{\lambda} = {}^nC_0 {}^nC_1 - {}^nC_1 {}^nC_2 + {}^nC_2 {}^nC_3 - ..... + (-1)^{n-2} \cdot {}^nC_{n-2} \cdot {}^nC_{n-1}$

$$\Rightarrow q = \begin{cases} n & ; \text{ if n is even} \\ {}^nC_{\frac{n-1}{2}} (-1)^{\frac{n-1}{2}} - (-1)^{n-1} n & ; \text{ if n is odd} \end{cases}$$

40. B, C

Sol. Clearly all lines are passing through origin, and if they are coplanar, then they are perpendicular to a line passing through origin and with direction cosines (ℓ_1, m_1, n_1)

$\Rightarrow a\ell_1 + \beta m_1 + \gamma n_1 = 0$; $\ell\ell_1 + mm_1 + nn_1 = 0$ and $3\alpha\ell_1 + 7\beta m_1 + 5\gamma n_1 = 0$

$$\Rightarrow \begin{vmatrix} \alpha & \beta & \gamma \\ \ell & m & n \\ 3\alpha & 7\beta & 5\gamma \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} 1 & 1 & 1 \\ \frac{\ell}{\alpha} & \frac{m}{\beta} & \frac{n}{\gamma} \\ 3 & 7 & 5 \end{vmatrix} = 0 \Rightarrow \frac{2\ell}{\alpha} + \frac{2m}{\beta} - \frac{4n}{\gamma} = 0$$

$$\Rightarrow \frac{\ell}{\alpha} + \frac{m}{\beta} = \frac{2n}{\gamma} \Rightarrow k_1 = 1 \text{ and } k_2 = 2$$

41. A, D

Sol. If $\vec{a}, \vec{b}, \vec{c}$ are the position vectors of A, B, C, then position vector of circumcentre of tetrahedron is given by $\vec{r} = \dfrac{c^2(\vec{a} \times \vec{b}) + b^2(\vec{c} \times \vec{a}) + a^2(\vec{a} \times \vec{b})}{2[\vec{a}\ \vec{b}\ \vec{c}]}$

We have $|\vec{r} - \vec{a}| = |\vec{r} - \vec{b}| = |\vec{r} - \vec{c}| = |\vec{r}|$

Consider $|\vec{r} - \vec{a}| = |\vec{r}| \Rightarrow a^2 = 2\vec{a} \cdot \vec{r}$ similarly, $b^2 = 2\vec{b} \cdot \vec{r}$ and $c^2 = 2\vec{c} \cdot \vec{r}$

Let $\vec{r} = x(\vec{b} \times \vec{c}) + y(\vec{c} \times \vec{a}) + z(\vec{a} \times \vec{b}) \Rightarrow \vec{r} \cdot \vec{a} = x[\vec{a}\ \vec{b}\ \vec{c}]$

$$\Rightarrow x = \frac{\vec{r} \cdot \vec{a}}{[\vec{a}\ \vec{b}\ \vec{c}]},\ y = \frac{\vec{r} \cdot \vec{b}}{[\vec{a}\ \vec{b}\ \vec{c}]},\ z = \frac{\vec{r} \cdot \vec{c}}{[\vec{a}\ \vec{b}\ \vec{c}]}$$

SECTION – B

42. 34

Sol. Let P_n be number of ways to cover $2 \times n$ board, then $P_n = P_{n-1} + P_{n-2}$

43. 32

Sol. Since term is independent of x, $\alpha = 15$

$\Rightarrow$ G.T. = ${}^{30}C_{15} 2^{-15}$

$\Rightarrow \alpha = 15, \beta = 2, \gamma = 15$

44. 2025

Sol. $T_n = \dfrac{n^2 + n + 1}{n(n+1)}$

45. 870

Sol. Sum (required) = $\sum_{i=1}^{12}\left[(i-1)12+j_i\right]$ = 12(0 + 1 + ... + 11) + (1 + 2 + 3 + ... + 12) = 870

46. 3

Sol. Here, $z_1z_2 + z_2z_3 + z_3z_1 = z_1z_2z_3\left(\bar{z}_1+\bar{z}_2+\bar{z}_3\right)$

Let $z = z_1 + z_2 + z_3 \Rightarrow |z|^3 = |3|z|^2 - 4| \Rightarrow |z| = 1$ or 2

47. 1214

Sol. Probability required = $\dfrac{{}^6C_5\times{}^8C_0+{}^8C_4\times{}^6C_1}{{}^{14}C_5}=\dfrac{213}{1001}$

SECTION – C

48. 8.96

49. 2.72

Sol. (Q.48.-49.):

Here, $f(x) = (x + 1)5^{-x}$

$\Rightarrow P(X = x) = k(x + 1)5^{-x}$

$\sum_{x=0}^{\infty}P(X=x)=1 \Rightarrow k=\dfrac{16}{25}$

$\Rightarrow P(x \le 1) = \dfrac{7k}{25}=\dfrac{112}{125}$ and $P(X \ge 3) = 1-\dfrac{38}{25}k=\dfrac{17}{625}$

50. 665.00

51. 1.60

Sol. (Q.50.-51.):

Here, $g(k) = \dfrac{1}{2}\left[(2k+1)^{\frac{3}{2}}-(2k-1)^{\frac{3}{2}}\right]$